Yocto Project and OpenEmbedded: Recent Changes and Future Directions

Embedded Recipes 2025

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Corrections, suggestions, contributions and translations are welcome!

Buildroot community: support and contribution

Documentation

- Buildroot comes with its own documentation
- Pre-built versions available at `https://buildroot.org/docs.html` (PDF, HTML, text)
- Source code of the manual located in `docs/manual` in the Buildroot sources
  - Written in *Asciidoc* format
- The manual can be built with:
  - `make manual`
  - or just `make manual-html`, `make manual-pdf`,

```
make
manual-epub
```

, `make manual-text`, `make manual-split-html`

Documentation

- A number of tools need to be installed on your machine, see the manual itself.

Getting support

- Free support
  - The *mailing list* for e-mail discussion
    `http://lists.busybox.net/mailman/listinfo/buildroot`
    1400+ subscribers, quite heavy traffic.
  - The IRC channel, `#buildroot` on the OFTC network, for interactive discussion
    60+ people, most available during European daylight hours
  - Bug tracker
    `https://bugs.busybox.net/buglist.cgi?product=buildroot`
- Commercial support
  - A number of embedded Linux services companies, including Bootlin, can provide commercial services around Buildroot.

Tips to get free support

- If you have a build issue to report:
  - Make sure to reproduce after a `make clean all` cycle
  - Include the Buildroot version, Buildroot `.config` that reproduces the issue, and last 100-200 lines of the build output in your report.
  - Use *pastebin* sites like `https://paste.ack.tf/` when reporting issues over IRC.
- The community will be much more likely to help you if you use a recent Buildroot version.

Release schedule

- The Buildroot community publishes stable releases every three months.
- YYYY.02, YYYY.05, YYYY.08 and YYYY.11 every year.
- The three months cycle is split in two periods
  - Two first months of active development
  - One month of stabilization before the release
- At the beginning of the stabilization phase, `-rc1` is released.
- Several `-rc` versions are published during this stabilization phase, until the final release.
- Development not completely stopped during the stabilization, a `next` branch is opened.
- Long-term maintenance of YYYY.02, with a *LTS initiative*

Contribution process

- Contributions are made in the form of patches
- Created with `git` and sent by e-mail to the mailing list
  - Use `git send-email` to avoid issues
  - Use `get-developers` to know to who patches should be sent
- The patches are reviewed, tested and discussed by the community
  - You may be requested to modify your patches, and submit updated versions
- Once ready, they are applied by one of the project maintainers
- Some contributions may be rejected if they do not fall within the Buildroot principles/ideas, as discussed by the community.

Patchwork

- Tool that records all patches sent on the mailing list
- Allows the community to see which patches need review/testing, and the maintainers which patches can be applied.
- Everyone can create an account to manage his own patches
- https://patchwork.ozlabs.org/project/buildroot/list/

ID	Patch	Series	A/F/R/T	S/W/F	Date	Submitter	Delegate	State
1059249	[1/1] package/bird: add optional protocols	[1/1] package/bird: add optional protocols	- - - -	0 0 0	2019-03-20	Adrien Gallouët		New
1059199	[v2,3/3] gummiboot: fix compatibility with newer glibc versions	Gummiboot fixup	- - - -	0 0 0	2019-03-20	Esben Haabendal		New
1059198	[v2,2/3] gummiboot: upgrade to last commit before being removed	Gummiboot fixup	- - - -	0 0 0	2019-03-20	Esben Haabendal		New
1059200	[v2,1/3] gummiboot: use new official upstream git repository	Gummiboot fixup	- - - -	0 0 0	2019-03-20	Esben Haabendal		New
1059145	[v2] support/testing: add xserver + Mesa OpenGL (GLX) + glxinfo	[v2] support/testing: add xserver + Mesa OpenGL (GLX) + glxinfo	- - - -	0 0 0	2019-03-20	Romain Naour		New
1058782	[v5,3/4] configs: qemu_riscv*_virt: Use OpenSBI by default	Untitled series #98112	- - - -	0 0 0	2019-03-20	Alistair Francis		New
1058784	[v5,4/4] boot/riscv-pk: Deprecate riscv-pk and BBL	[v5,1/4] board/qemu/riscv32-virt: Update linux config	- - - -	0 0 0	2019-03-20	Alistair Francis		New
1058785	[v5,2/4] boot/opensbi: new package	[v5,1/4] board/qemu/riscv32-virt: Update linux config	- - - -	0 0 0	2019-03-20	Alistair Francis		New
1058783	[v5,1/4] board/qemu/riscv32-virt: Update linux config	[v5,1/4] board/qemu/riscv32-virt: Update linux config	- - - -	0 0 0	2019-03-20	Alistair Francis		New
1058701	[v3,2/2] package/uftp: openssl is optional, not mandatory	[v3,1/2] package/uftp: fix openssl static linking	- - - -	0 0 0	2019-03-19	Fabrice Fontaine		New
1058700	[v3,1/2] package/uftp: fix openssl static linking	[v3,1/2] package/uftp: fix openssl static linking	- 1 - -	0 0 0	2019-03-19	Fabrice Fontaine		New
1058645	[1/1] mender-grubenv: new package	[1/1] mender-grubenv: new package	- - - -	0 0 0	2019-03-19	Adam Duskett		New
1058365	[RFC] openssh: add option to allow login as root	[RFC] openssh: add option to allow login as root	- - - -	0 0 0	2019-03-19	Esben Haabendal		New
1058357	[3/3] gummiboot: fix compatibility with newer glibc versions	Gummiboot fixup	- - - -	0 0 0	2019-03-19	Esben Haabendal		New
1058356	[2/3] gummiboot: upgrade to last commit before being removed	Gummiboot fixup	- - - -	0 0 0	2019-03-19	Esben Haabendal		New
1058355	[1/3] gummiboot: use new official upstream git repository	Gummiboot fixup	- - - -	0 0 0	2019-03-19	Esben Haabendal		New
1058140	[v2,8/8] support/testing: test_optee.py: test optee boot and testsuite	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058139	[v2,7/8] testing: test can use the locally generated qemu host tool	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058141	[v2,6/8] configs/qemu_armv7a_tz_virt: Armv7-A emulation with TrustZone services	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058136	[v2,5/8] package/optee-test: fix dependency in TAs build	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058137	[v2,4/8] boot/arm-trusted-firmware: support alternate image files	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058138	[v2,3/8] boot/arm-trusted-firmware: support debug mode	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058135	[v2,2/8] boot/arm-trusted-firmware: in-tree and OP-TEE BL32	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New
1058134	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	[v2,1/8] boot/arm-trusted-firmware: support 32bit Arm targets	- - - -	0 0 0	2019-03-18	Etienne Carriere		New

Automated build testing

- The enormous number of configuration options in Buildroot make it very difficult to test all combinations.
- Random configurations are therefore built 24/7 by multiple machines.
  - Random choice of architecture/toolchain combination from a pre-defined list
  - Random selection of packages using `make randpackageconfig`
  - Random enabling of features like static library only, or `BR2\_ENABLE\_DEBUG\=y`
- Scripts and tools publicly available at `https://gitlab.com/buildroot.org/buildroot-test/`
- Results visible at `http://autobuild.buildroot.org/`
- Daily e-mails with the build results of the past day

autobuild.buildroot.org

Buildroot tests

Date	Duration	Status	Commit ID	Submitter	Arch/Subarch	Failure reason	Libc	Static?	Data
2019-03-20 16:32:05	01:12:43	OK	master 83f7a923	Thomas Petazzoni (gcc159)	mips64el / mips64r6	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:30:42	04:02:55	OK	master 3033e83d	Thomas Petazzoni (gcc159)	arm / cortex-a8	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:30:27	25:32	OK	master f389df23	Yann E. MORIN	powerpc64le / power8	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:22:37	02:12:40	OK	master 3033e83d	Thomas Petazzoni (gcc160)	arc / archs	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:20:16	30:31	OK	master 83f7a923	Mark Corbin (Embecosm Godzilla U18.10)	riscv64	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:15:11	03:08:12	NOK	master 3033e83d	Yann E. MORIN	m68k / 68040	host-uboot-tools-2019.01	uclibc	N	dir, end log, config, defconfig
2019-03-20 16:05:02	02:47:01	NOK	master 3033e83d	Peter Korsgaard (gcc112/ppc64le/Centos 7/gcc 4.8.5)	arc / arc700	host-uboot-tools-2019.01	uclibc	N	dir, end log, config, defconfig
2019-03-20 16:04:03	31:27	OK	master 83f7a923	Yann E. MORIN	nios2	none	glibc	N	dir, end log, config, defconfig
2019-03-20 16:03:52	06:29:57	OK	master 3033e83d	Giulio Benetti (Micronova srl Server)	mipsel / mips32	none	uclibc	N	dir, end log, config, defconfig
2019-03-20 16:01:04	01:24:17	OK	master 3033e83d	Thomas Petazzoni (gcc159)	arc / archs	none	glibc	N	dir, end log, config, defconfig
2019-03-20 15:57:47	03:07:12	OK	master 3033e83d	Matt Weber (U14.04 Sandboxed)	xtensa	none	uclibc	N	dir, end log, config, defconfig
2019-03-20 15:50:20	22:35	OK	master 83f7a923	Andre Hentschel	arm / arm1176jzf-s	none	uclibc	N	dir, end log, config, defconfig
2019-03-20 15:49:56	01:13:49	NOK	master 3033e83d	Matt Weber (U14.04 Sandboxed)	arm / arm926ej-s	host-go-1.12.1	uclibc	Y	dir, end log, config, defconfig
2019-03-20 15:48:49	04:06:01	OK	master 3033e83d	Mark Corbin (Embecosm Godzilla U18.10)	riscv32	none	glibc	N	dir, end log, config, defconfig
2019-03-20 15:43:42	16:59	OK	2018.02.x 3a2c33cf	Thomas Petazzoni (gcc160)	powerpc / 603e	none	uclibc	N	dir, end log, config, defconfig
2019-03-20 15:37:11	56:16	OK	master 3033e83d	Yann E. MORIN	nios2	none	glibc	N	dir, end log, config, defconfig
2019-03-20 15:31:34	01:52:13	OK	2018.11.x 3e37abb3	Yann E. MORIN	xtensa	none	uclibc	N	dir, end log, config, defconfig
2019-03-20 15:26:00	01:02:47	OK	2018.02.x 3a2c33cf	Thomas Petazzoni (gcc160)	arm / arm926ej-s	none	uclibc	Y	dir, end log, config, defconfig
2019-03-20 15:25:55	01:46:38	OK	master 3033e83d	Andre Hentschel	mips64el / mips64r6	none	glibc	N	dir, end log, config, defconfig

[fragile]Autobuild daily reports

```
Subject: [Buildroot] [autobuild.buildroot.net] Build results for 2019-03-19

Build statistics for 2019-03-19
===============================

      branch |  OK | NOK | TIM | TOT |
   2018.02.x |  18 |   3 |   0 |  21 |
   2018.11.x |  36 |   1 |   0 |  37 |
   2019.02.x |  25 |   4 |   0 |  29 |
      master | 166 | 105 |   3 | 274 |

Results for branch 'master'
===========================

Classification of failures by reason
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```

[fragile]Autobuild daily reports

```
                        unknown | 22
          angularjs-legal-info | 15
      host-uboot-tools-2019.01 | 11
[...]

Detail of failures
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       sparc | android-tools-4.2.2+git2013... | NOK | http://autobuild.
buildroot.net/results/f1648f245d77f85661bc0d2f1e8097c3695206d8 |
    mips64el |           angularjs-legal-info | NOK | http://autobuild.
buildroot.net/results/fdf6b64648dfa58ec74de31104a1a71248242d80 |
[...]
         arm |         glib-networking-2.58.0 | NOK | http://autobuild.
buildroot.net/results/fc2e68921bd84d13d2e9bc900a91e46b08d698fe |
```

Additional testing effort

- Run-time test infrastructure in `support/testing`
  - Contains a number of test cases that verify that specific Buildroot configurations build correctly, and boot correctly under Qemu.
  - Validates filesystem format support, specific packages, core Buildroot functionality.
  - `./support/testing/run-tests -l`
  - `./support/testing/run-tests tests.fs.test\_ext.TestExt2`
  - Run regularly on *Gitlab CI*
- All *defconfigs* in `configs/` are built every week on *Gitlab CI*